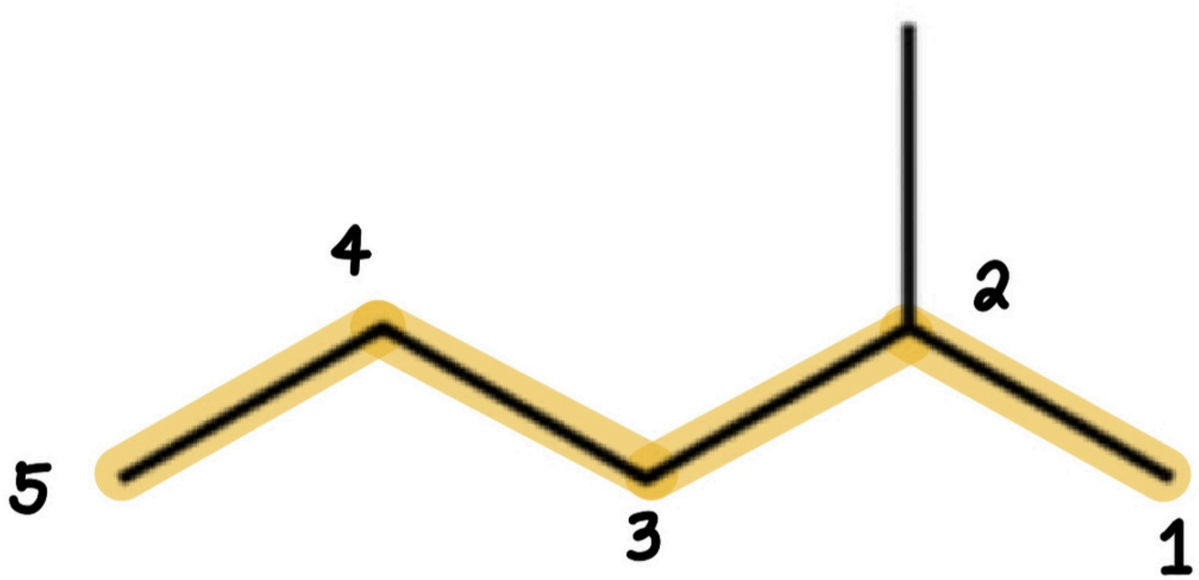
A01 (FOUNDATION): C_6H_{14}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 5 C (pentane)

Numbering: left-right gives 4; right-left gives 2

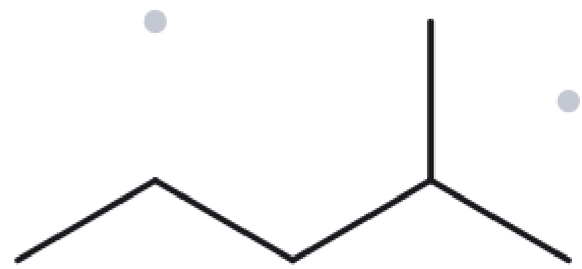
- First Point of Difference: 2 < 4, so number from the right

Substituents: methyl on C-2 → 2-methylpentane

3. My Name

2-methylpentane

4. The Alkane Check



2-methylpentane

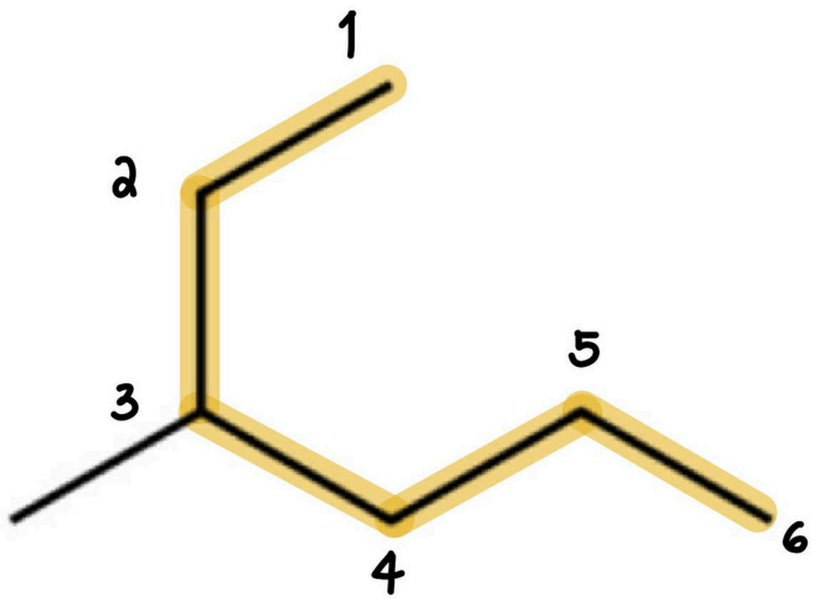
Returned Name: 2-methylpentane

Check: MATCH

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A02 (FOUNDATION): C_7H_{16}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 6 C (hexane)

Numbering: left-right gives 3; right-left gives 4

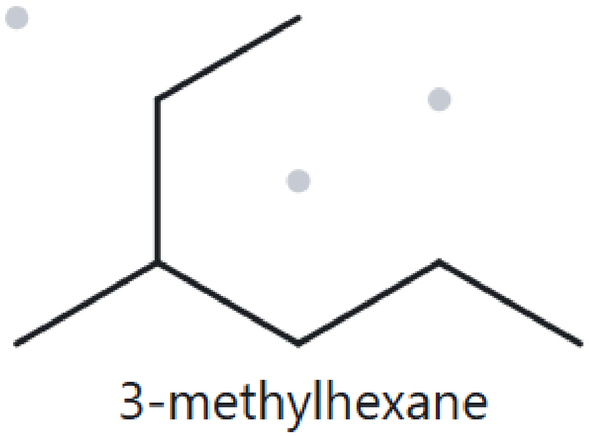
- First Point of Difference: 3 < 4, so number from the left

Substituents: methyl on C-3 → 3-methylhexane

3. My Name

3-methylhexane

4. The Alkane Check



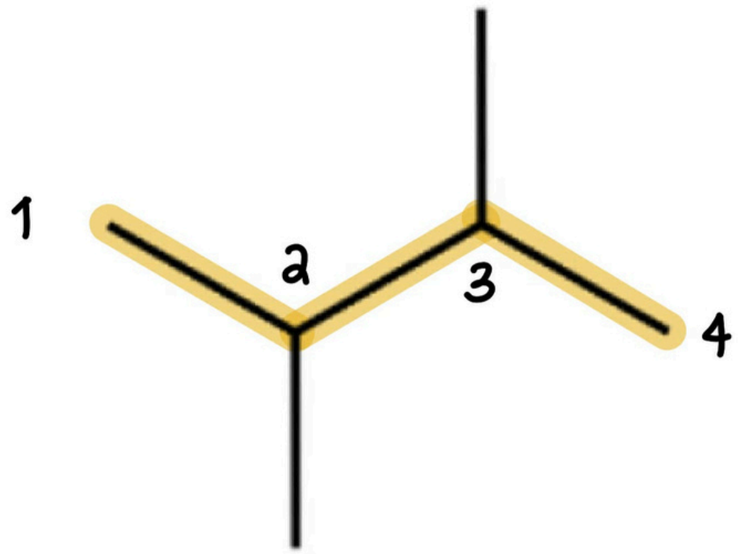
Returned Name: 3-methylhexane

Check: MATCH

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A03 (FOUNDATION): C_6H_{14}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 4 C (butane)

Numbering: left-right and right-left give the same locants so either direction is correct.

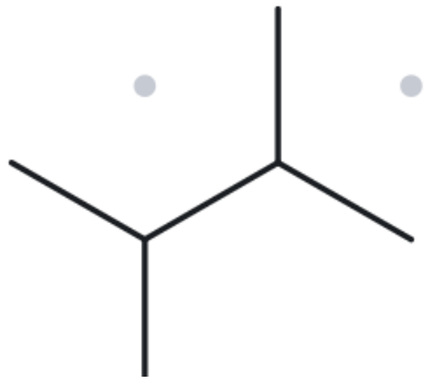
Substituents: methyl on C-2 and C-3

→ 2,3-dimethylbutane

3. My Name

2,3-dimethylbutane

4. The Alkane Check



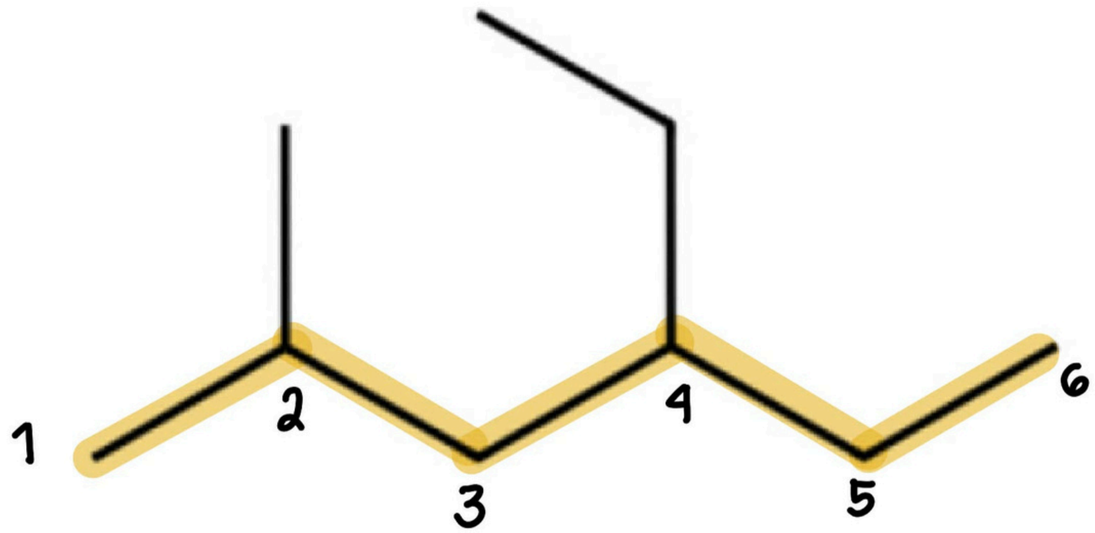
Returned Name: 2,3-dimethylbutane

Check: MATCH

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A04 (DEVELOPING): C_9H_{20}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 6 C (hexane)

Numbering: left-right gives 2,4 ; right-left gives 3,5

- First Point of Difference: 2 < 3, so number from the left

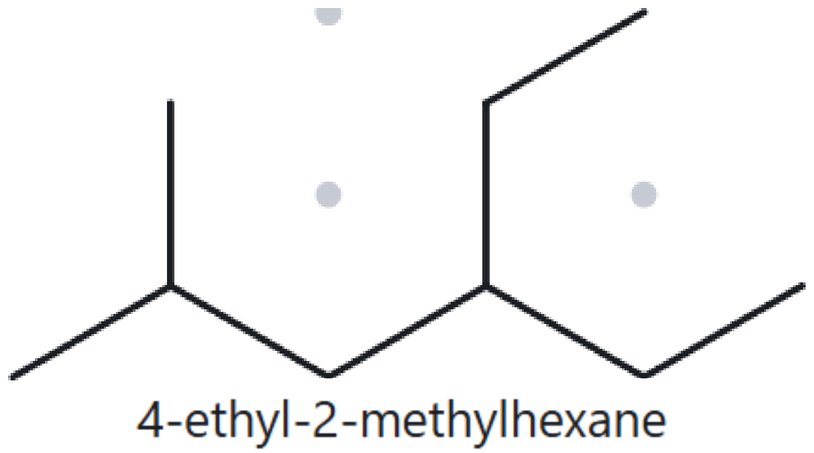
Substituents: methyl on C-2, ethyl on C-4

→ 4-ethyl-2-methylhexane

3. My Name

4-ethyl-2-methylhexane

4. The Alkane Check



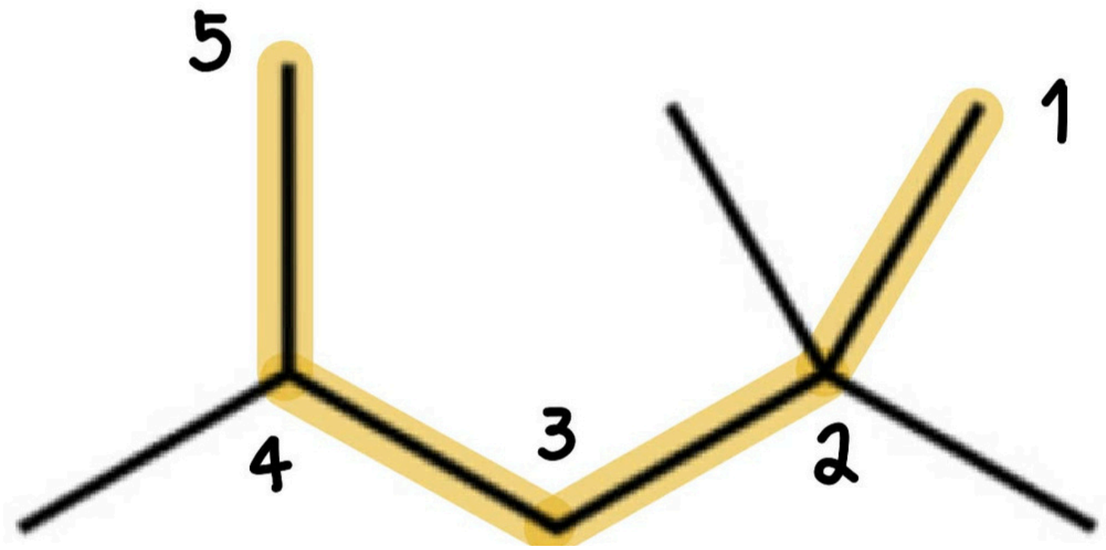
Returned Name: 4-ethyl-2-methylhexane

Check: MATCH

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A05 (DEVELOPING): C_8H_{18}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 5 C (pentane)

Numbering: left-right gives 2-4-4 ; right-left gives 2-2-4

- First Point of Difference: 2 < 4, so number from the right

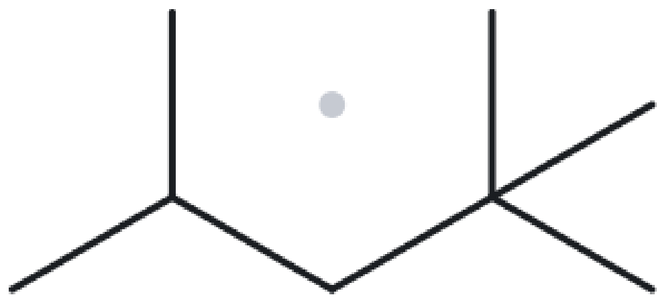
Substituents: methyl on C-4 and 2 methyls on C-2

→ 2,2,4-trimethylpentane

3. My Name

2,2,4-trimethylpentane

4. The Alkane Check



2,2,4-trimethylpentane

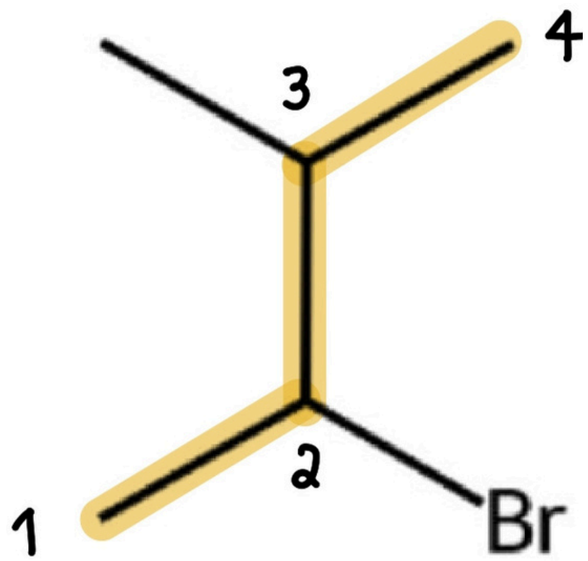
Returned Name: 2,2,4-trimethylpentane

Check: MATCH

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A06 (DEVELOPING): $C_5H_{11}Br$

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 4 C (butane)

Numbering: left-right gives 2,3; right-left gives 2,3

- Tie-breaker: Locant sets are identical (2,3), so we number alphabetically (bromo before methyl). Number from left to give bromo the lower number.

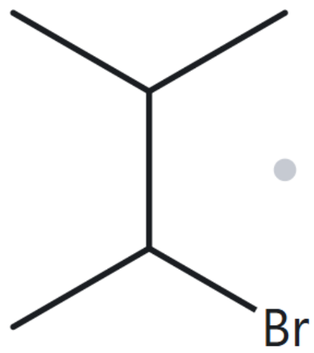
Substituents: bromo on C-2 and methyl on C-3

→ 2-bromo-3-methylbutane

3. My Name

2-bromo-3-methylbutane

4. The Alkane Check



2-bromo-3-methylbutane

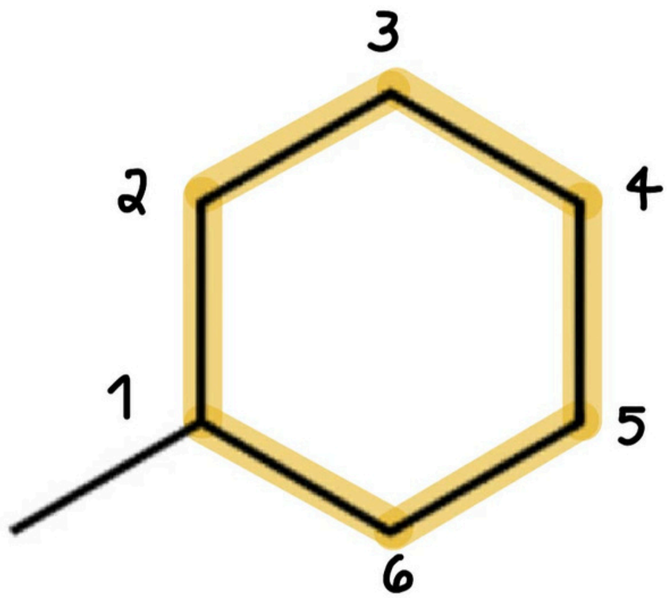
Returned Name: 2-bromo-3-methylbutane

Check: MATCH

- ✓ longest chain confirmed
- ✓ lowest locants
- ✓ alphabetical order
- ✓ punctuation
- ✓ formula counted

A07 (DEVELOPING): C_7H_{14}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 6 C (cyclohexane)

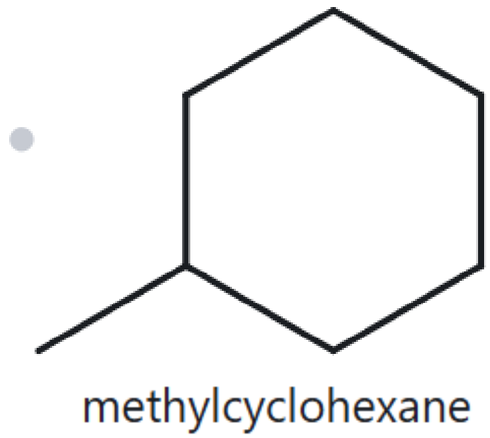
Numbering: Start at the substituent to give it the lowest possible numbering (C-1)

Substituents: Methyl on C-1 → methylcyclohexane

3. My Name

methylcyclohexane

4. The Alkane Check



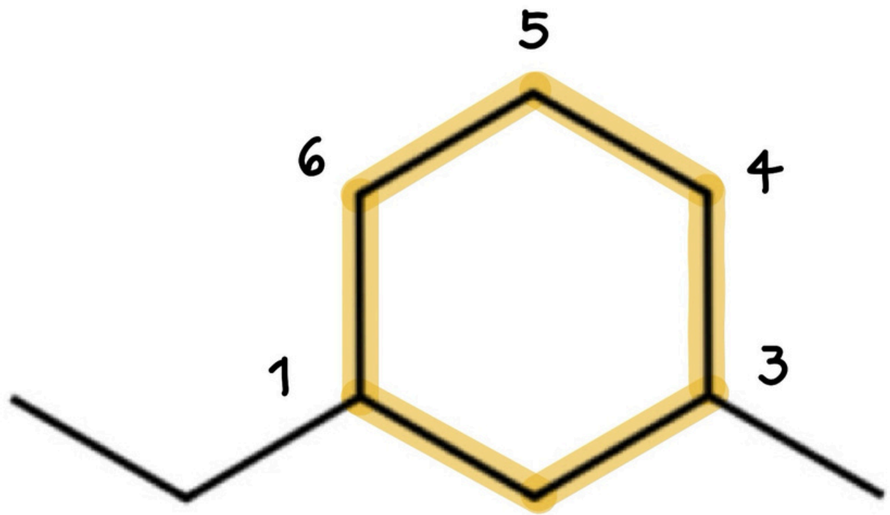
Returned Name: methylcyclohexane

Check: MATCH

- ✓ longest chain confirmed
- ✓ lowest locants
- ✓ alphabetical order
- ✓ punctuation
- ✓ formula counted

A08 (CHALLENGE): C_9H_{18}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 6 C (cyclohexane)

Numbering: Clockwise gives 1,3; Counterclockwise gives 1,3

- Tie-breaker: Locant sets are identical (1,3), so we number alphabetically (ethyl before methyl). Number counterclockwise to give ethyl the lower number.

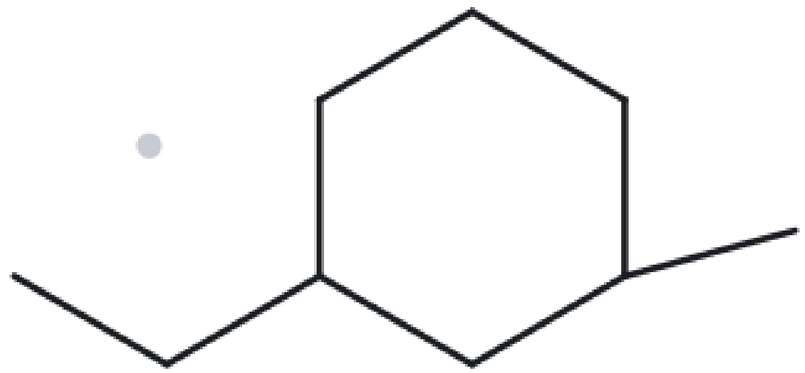
Substituents: Ethyl on C-1 and methyl on C-3

→ 1-ethyl-3-methylcyclohexane

3. My Name

1-ethyl-3-methylcyclohexane

4. The Alkane Check



1-ethyl-3-methylcyclohexane

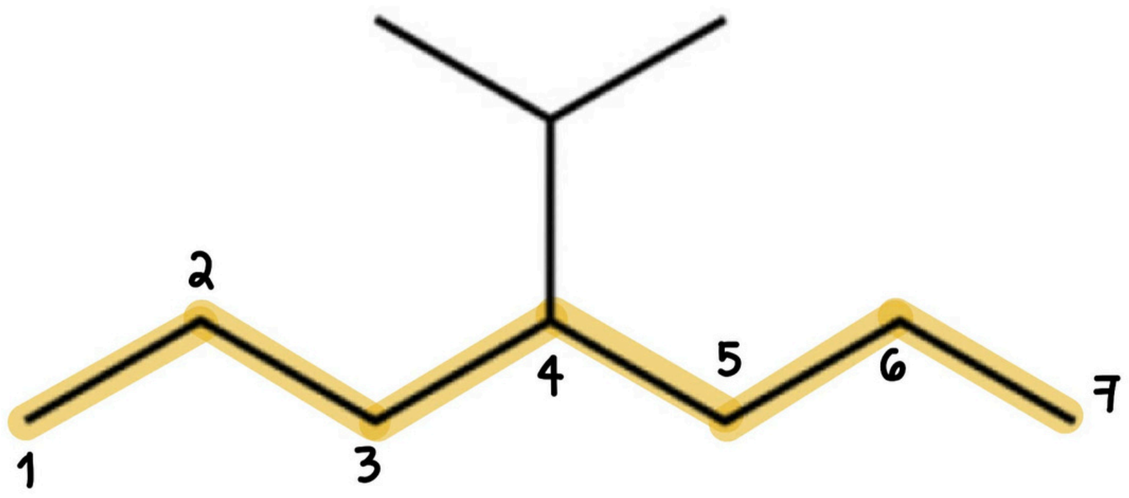
Returned Name: 1-ethyl-3-methylcyclohexane

Check: MATCH

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A09 (CHALLENGE): $C_{10}H_{22}$

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 7 C (heptane)

Numbering: left-right gives 4; right-left gives 4

- Tie-breaker: Locant sets are identical (4)

Substituents: Isopropyl on C-4

→ 4-isopropylheptane

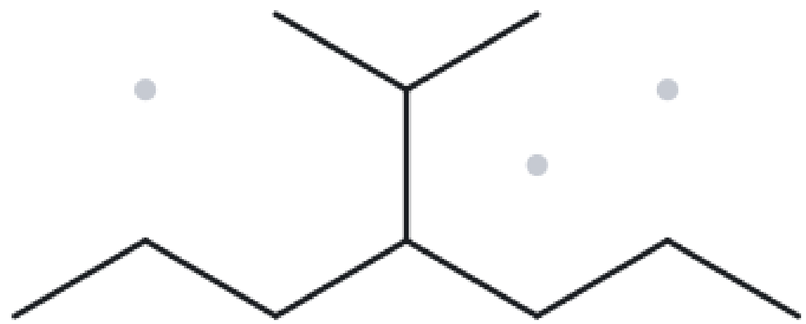
Alkane Check: (Additional)

- 4-isopropylheptane is the correct name because of the Principal Chain Rule

3. My Name

4-isopropylheptane

4. The Alkane Check



2-methyl-3-propylhexane

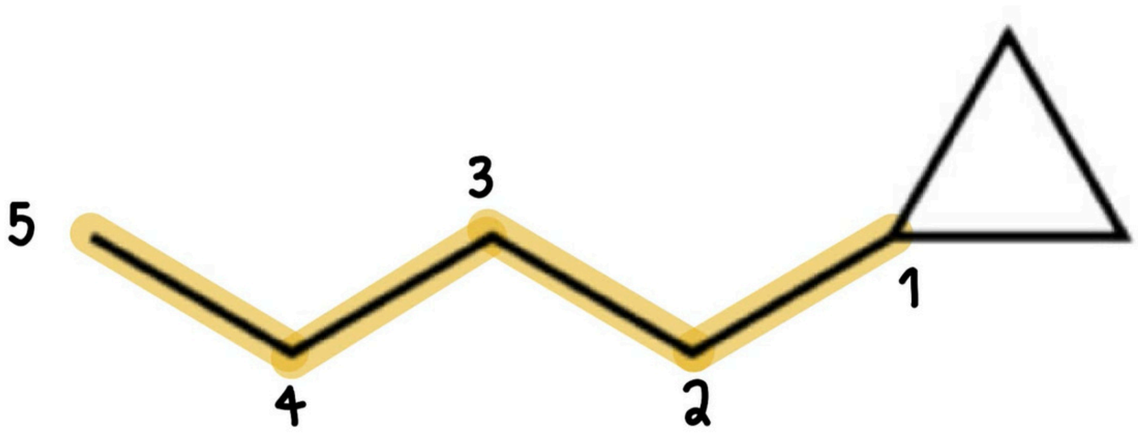
Returned Name: 2-methyl-3-propylhexane

Check: MISMATCH

✅ longest chain confirmed ✅ lowest locants ✅ alphabetical order ✅ punctuation ✅ formula counted

A10 (CHALLENGE): C_8H_{16}

1. Numbered Parent



2. Naming Trail

Parent: longest chain = 5 C (pentane)

Numbering: left-right gives 5; right-left gives 1

- First Point of Difference: 1 < 5, we number the parent chain from the right

Substituents: Cyclopropyl ring on C-1

→ 1-cyclopropylpentane

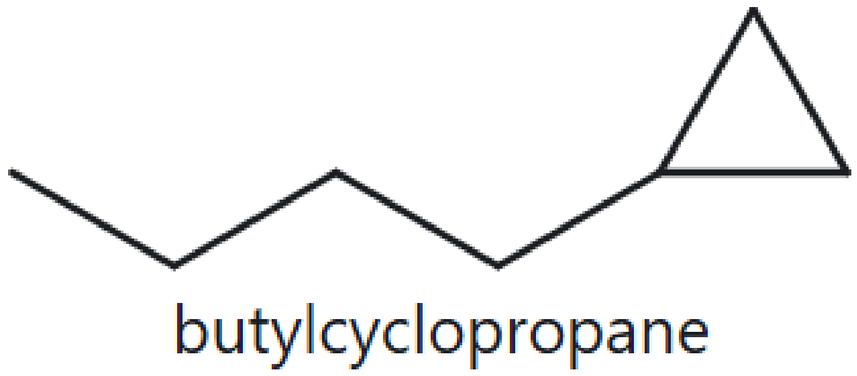
Alkane Check: (Additional)

- 1-cyclopropylpentane is the correct name because of the Principal Chain Rule

3. My Name

1-cyclopropylpentane

4. The Alkane Check



Returned Name: butylcyclopropane

Check: MISMATCH

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

B01 (FOUNDATION): 2-methylpropane

1. Name Decoded

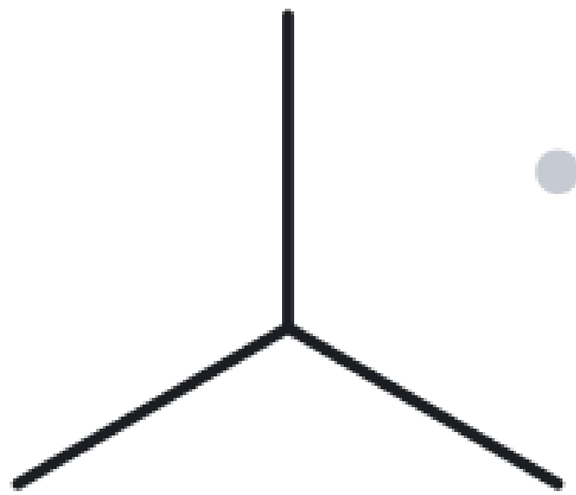
Parent: propane → a 3-carbon chain

Substituent: Methyl on C-2

Expected formula: 3 + 1 = 4 C; 8 + 2 = 10 H

→ C_4H_{10}

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_4H_{10}

Propane: C_3H_8

Methyl: CH_3

Minus one H for CH_3 = C_4H_{10} ✓

4. Back-check

Returned by the Alkane:

2-methylpropane

Check: Same as the given name

- ☑ parent length matches
- ☑ substituents positions match
- ☑ formula agrees
- ☑ back-check recorded

B02 (FOUNDATION): 3-ethylpentane

1. Name Decoded

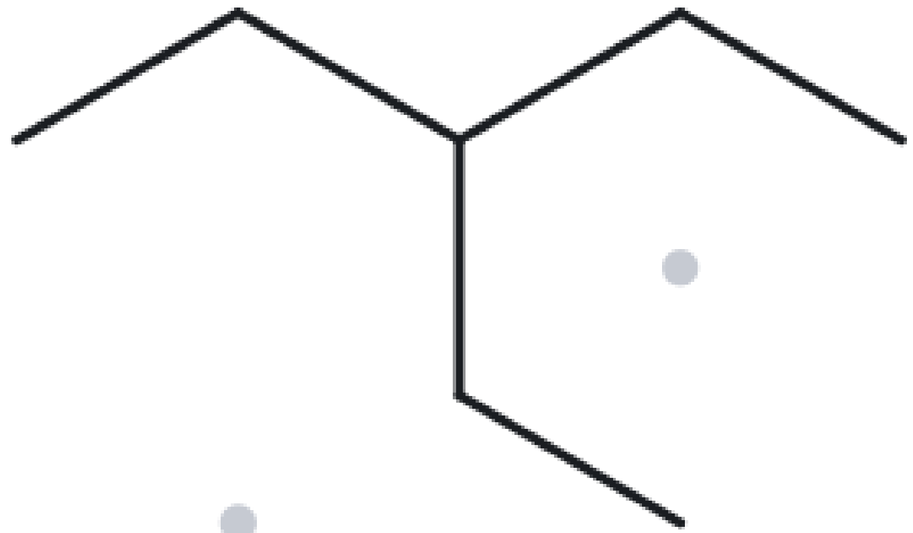
Parent: pentane → a 5-carbon chain

Substituent: Ethyl on C-3

Expected formula: 5 + 2 = 7 C; 12 + 4 = 16 H

→ C_7H_{16}

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_7H_{16}

Pentane: C_5H_{12}

Ethyl: C_2H_5

Minus one H for C_2H_5 = C_7H_{16} ✓

4. Back-check

Returned by the Alkane:

3-ethylpentane

Check: Same as the given name

- ☑ parent length matches
- ☑ substituents positions match
- ☑ formula agrees
- ☑ back-check recorded

B03 (FOUNDATION): 2,2-dimethylbutane

1. Name Decoded

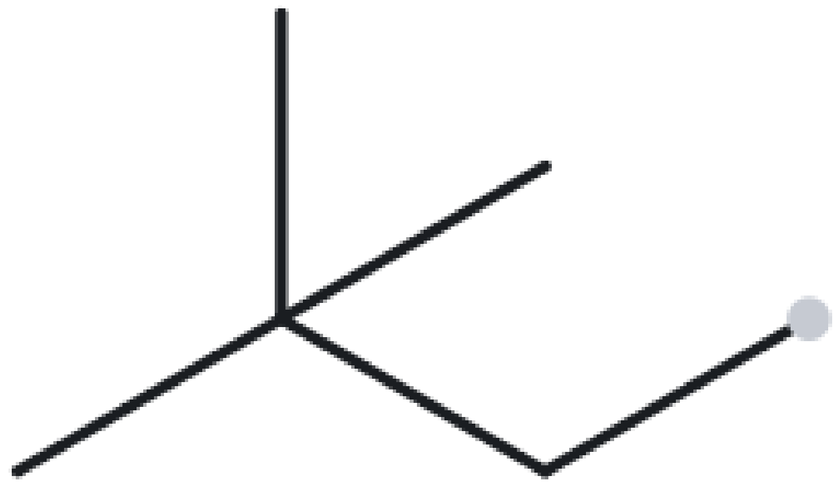
Parent: butane → a 4-carbon chain

Substituent: 2 Methyl on C-2

Expected formula: 4 + 2 = 6 C; 10 + 4 = 14 H

→ C_6H_{14}

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_6H_{14}

Butane: C_4H_{10}

Methyl: 2 CH_3

Minus one H for each CH_3 = C_6H_{14} ✓

4. Back-check

Returned by the Alkane:

2,2-dimethylbutane

Check: Same as the given name

- ☑ parent length matches
- ☑ substituents positions match
- ☑ formula agrees
- ☑ back-check recorded

B04 (DEVELOPING): 3-ethyl-2-methylpentane

1. Name Decoded

Parent: pentane → a 5-carbon chain

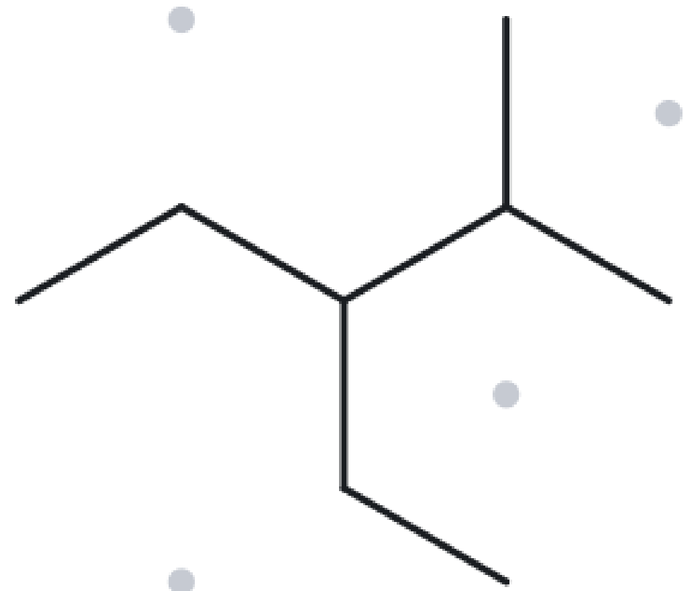
Substituent: Ethyl on C-3, Methyl on C-2

Expected formula: 5 + 2 + 1 = 8 C;

12 + 4 + 2 = 18 H

→ C_8H_{18}

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_8H_{18}

Pentane: C_5H_{12}

Methyl: CH_3

Ethyl: C_2H_5

Minus one H for each CH_3 and C_2H_5 = C_8H_{18} ✓

4. Back-check

Returned by the Alkane:

3-ethyl-2-methylpentane

Check: Same as the given name

- ☑ parent length matches
- ☑ substituents positions match
- ☑ formula agrees
- ☑ back-check recorded

B05 (DEVELOPING): 1-chloro-2-methylpropane

1. Name Decoded

Parent: propane → a 3-carbon chain

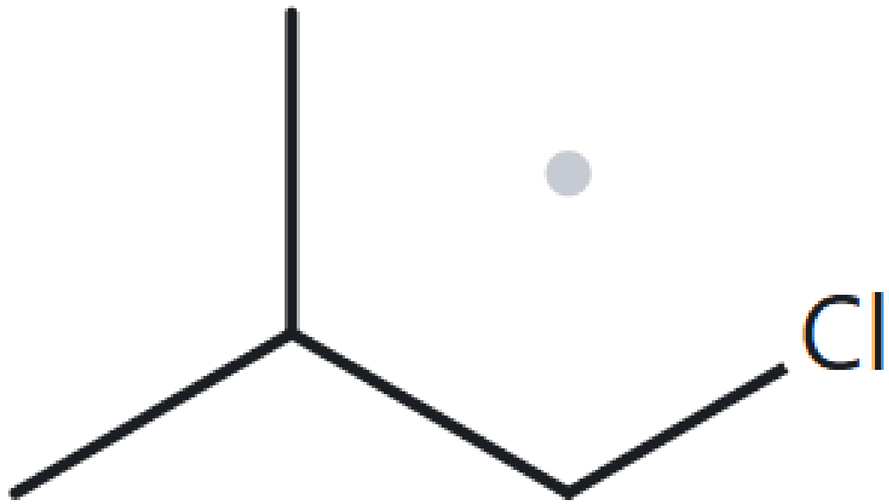
Substituent: Chloro on C-1; Methyl on C-2

Expected formula: 1 Cl; 3 + 1 = 4 C;

8 + 1 = 9 H

→ C_4H_9Cl

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_4H_9Cl

Propane: C_3H_8

Methyl: CH_3

Minus two H for CH_3 due to presence of Cl

= C_4H_9Cl ✓

4. Back-check

Returned by the Alkane:

1-chloro-2-methylpropane

Check: Same as the given name

- ☑ parent length matches
- ☑ substituents positions match
- ☑ formula agrees
- ☑ back-check recorded

B06 (DEVELOPING): 1,2-dimethylcyclopentane

1. Name Decoded

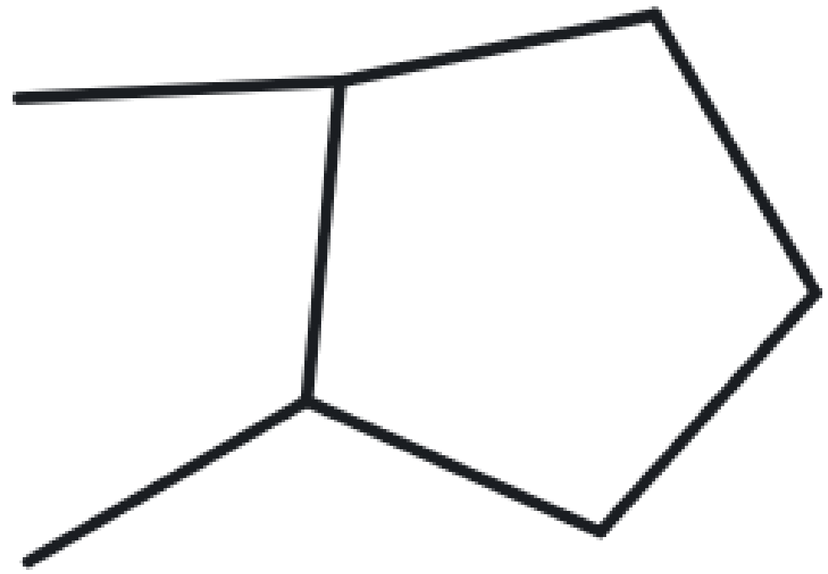
Parent: cyclopentane → a 5-carbon ring

Substituent: Methyl on C-1 and C-2

Expected formula: 5 + 2 = 7 C; 10 + 4 = 14 H

→ C_7H_{14}

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the drawing: C_7H_{14}

Cyclopentane: C_5H_{10}

Methyl: 2 CH_3

Minus H for each CH_3= C_7H_{14} ✓

4. Back-check

Returned by the Alkane:

1,2-dimethylcyclopentane

Check: Same as the given name

- ☑ parent length matches
- ☑ substituents positions match
- ☑ formula agrees
- ☑ back-check recorded

B07 (DEVELOPING): 4-ethyl-3,3-dimethylheptane

1. Name Decoded

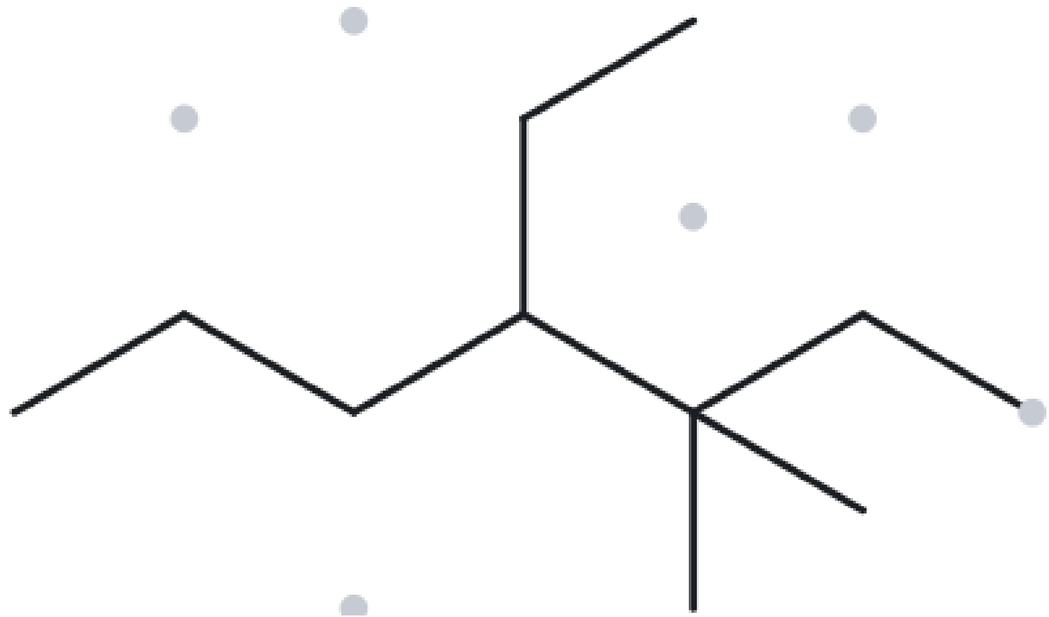
Parent: heptane → a 7-carbon chain

Substituent: Ethyl on C-4; 2 Methyl on C-3

Expected formula: 7 + 2 + 2 = 11 C

→ $C_{11}H_{24}$

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the drawing: $C_{11}H_{24}$

Heptane: C_7H_{16}

Ethyl: C_2H_5

Methyl: 2 CH_3

Minus H for each CH_3= $C_{11}H_{24}$ ✓

4. Back-check

Returned by the Alkane:

4-ethyl-3,3-dimethylheptane

Check: Same as the given name

✓ parent length matches ✓ substituents positions match ✓ formula agrees ✓ back-check recorded

B08 (CHALLENGE): tert-butylcyclohexane

1. Name Decoded

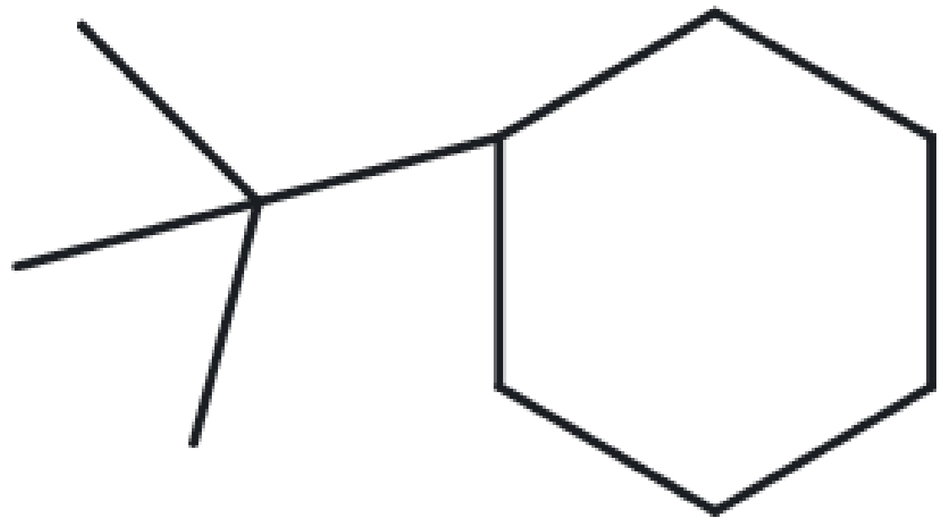
Parent: cyclohexane→ a 6-carbon ring

Substituent: Tert-butyl on C-1

Expected formula: 6+ 4 = 10 C

→ $C_{10}H_{20}$

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the drawing: $C_{10}H_{20}$

Cyclohexane: C_6H_{12}

Tert-butyl: C_4H_9

Minus H for connection= $C_{10}H_{20}$ ✓

4. Back-check

Returned by the Alkane:

(1,1-dimethylethyl)cyclohexane

Check: Same as the given name since tert-butyl and 1,1-dimethylethyl is the same

- ✓ parent length matches
- ✓ substituents positions match
- ✓ formula agrees
- ✓ back-check recorded

B09 (CHALLENGE): 3-ethyl-1,1-dimethylcyclohexane

1. Name Decoded

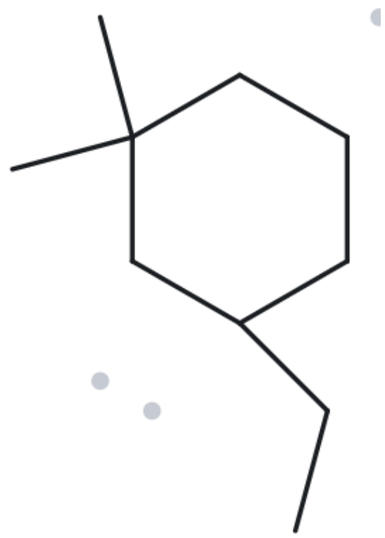
Parent: cyclohexane→ a 6-carbon ring

Substituent: 2 Methyl on C-1 and 1 Ethyl on C-3

Expected formula: 6 + 1 + 1 + 2 = 10 C

→ $C_{10}H_{20}$

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the drawing: $C_{10}H_{20}$

Cyclohexane: C_6H_{12}

Methyl: 2 CH_3

Ethyl: C_2H_5

Minus H for connections = $C_{10}H_{20}$ ✓

4. Back-check

Returned by the Alkane:

3-ethyl-1,1-dimethylcyclohexane

Check: Same as the given name

✓ parent length matches ✓ substituents positions match ✓ formula agrees ✓ back-check recorded

B10 (CHALLENGE): 4-ethyl-3,3-dimethylheptane

1. Name Decoded

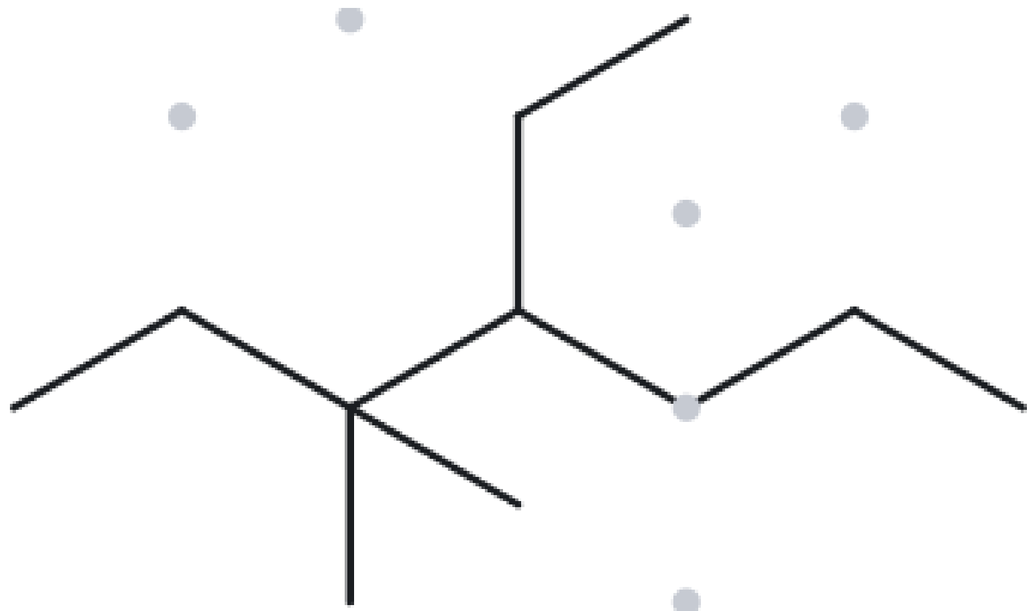
Parent: heptane→ a 7-carbon straight chain

Substituent: 2 Methyl on C-3; Ethyl on C-4

Expected formula: 7 + 1 + 1 + 2 = 11 C

→ $C_{11}H_{24}$

2. Bond-line Structure (The Alkane)



3. Formula Check

Counted from the drawing: $C_{11}H_{24}$

Heptane: C_7H_{16}

Methyl: 2 CH_3

Ethyl: C_2H_5

Minus H for connections = $C_{11}H_{24}$ ✓

4. Back-check

Returned by the Alkane:

4-ethyl-3,3-dimethylheptane

Check: Same as the given name

- ✓ parent length matches
- ✓ substituents positions match
- ✓ formula agrees
- ✓ back-check recorded